

CHAPTER 4

FORCES

1.5 Forces

1.5.1 Balanced and unbalanced forces

- 1 Identify and use different types of force, including weight (gravitational force), friction, drag, air resistance, tension (elastic force), electrostatic force, magnetic force, thrust (driving force) and contact force
- 2 Identify forces acting on an object and draw free-body diagram(s) representing the forces
- 3 State Newton's first law as 'an object either remains at rest or continues to move in a straight line at constant speed unless acted on by a resultant force'
- 4 State that a force may change the velocity of an object by changing its direction of motion or its speed
- 5 Determine the resultant of two or more forces acting along the same straight line
- 6 Recall and use the equation
resultant force = mass $\times$ acceleration
 $F = ma$
- 7 State Newton's third law as 'when object A exerts a force on object B, then object B exerts an equal and opposite force on object A'
- 8 Know that Newton's third law describes pairs of forces of the same type acting on different objects

1.5.2 Friction

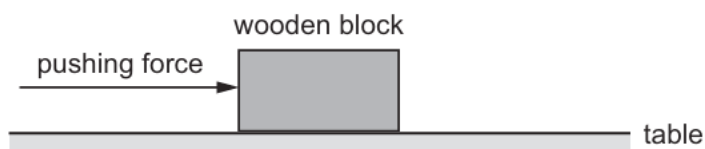
- 1 Describe friction as a force that may impede motion and produce heating
- 2 Understand the motion of objects acted on by a constant weight or driving force, with and without drag (including air resistance or resistance in a liquid)
- 3 Explain how an object reaches terminal velocity
- 4 Define the thinking distance, braking distance and stopping distance of a moving vehicle
- 5 Explain the factors that affect thinking and braking distance including speed, tiredness, alcohol, drugs, load, tyre surface and road conditions

1.5.4 Circular motion

- 1 Describe, qualitatively, motion in a circular path due to a force perpendicular to the motion as:
 - (a) speed increases if force increases, with mass and radius constant
 - (b) radius decreases if force increases, with mass and speed constant
 - (c) an increased mass requires an increased force to keep speed and radius constant
$$(F = \frac{mv^2}{r} \text{ is not required})$$

1. O/N/23/11/Q3

A wooden block is pushed across a table at constant speed.

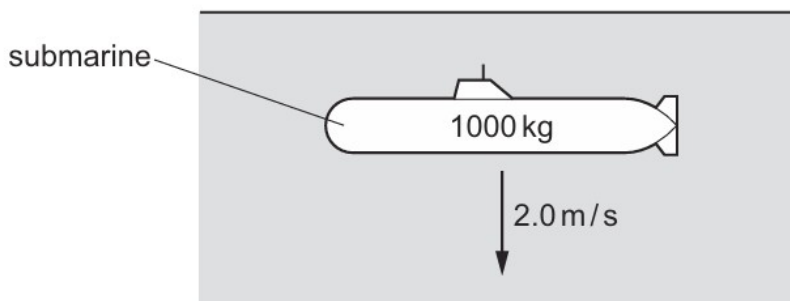


Which statement is correct?

- A** The frictional force increases as the block moves at constant speed.
- B** The frictional force is equal and opposite to the pushing force.
- C** The frictional force is greater than the pushing force.
- D** The frictional force is less than the pushing force.

2. O/N/23/12/Q5

A small submarine of mass 1000 kg sinks in water with a constant speed of 2.0 m/s.



What is the resultant force exerted on the submarine as it sinks?

- A** 0
- B** 500 N downwards
- C** 2000 N downwards
- D** 9800 N downwards

3. O/N/23/12/Q6

A student drops, from rest, a light ball in air. The ball experiences some air resistance as it falls.

What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

4. O/N/23/12/Q7

The stopping distance of a car travelling at a certain speed is measured in four different situations.

- on a dry road with new tyres
- on a dry road with old, worn tyres
- on a wet road with new tyres
- on a wet road with old, worn tyres

The results are shown.

	W	X	Y	Z
stopping distance / m	12	16	17	40

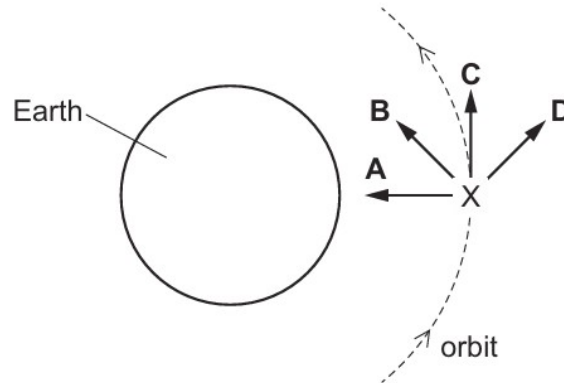
Which two results are possible measurements on the dry road with old, worn tyres?

- A** W and X **B** X and Y **C** X and Z **D** Y and Z

5. O/N/23/12/Q9

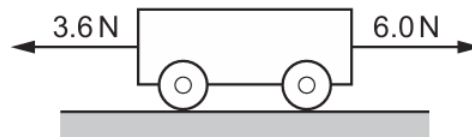
The International Space Station orbits the Earth.

Which arrow shows the direction of the force on the International Space Station when it is at point X?



6. M/J/23/11/Q7

The diagram shows the only two horizontal forces acting on a trolley that is accelerating at 0.12 m/s^2 .



What is the mass of the trolley?

- A** 0.050 kg **B** 0.13 kg **C** 20 kg **D** 80 kg

7. M/J/23/12/Q8

Newton's third law describes a pair of forces.

Which row shows whether the two forces are of the same type and whether they act on the same object?

	the types of forces in the pair	the objects on which they act
A	different	different
B	different	same
C	same	different
D	same	same

8. M/J/23/12/Q9

The stopping distance of a car is the sum of the thinking distance and the braking distance.

Which factors affect the braking distance?

	speed of car	tiredness of driver	condition of road
A	✓	✗	✗
B	✓	✓	✓
C	✓	✗	✓
D	✗	✓	✓

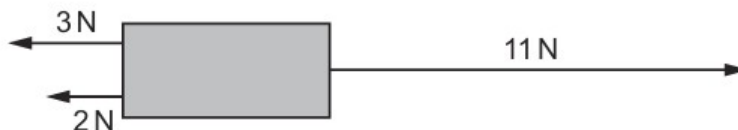
key

✓ = affects braking distance

✗ = does not affect braking distance

9. O/N/22/11/Q1

The diagram shows three forces acting on a block. The resultant force is 6 N to the right.



Which additional force produces a resultant force of 3 N to the left?

- A** 3 N to the left
- B** 9 N to the left
- C** 6 N to the right
- D** 13 N to the right

10. O/N/22/11/Q3

A car of mass 1000 kg is travelling down a steep hill. The brakes fail and the driver uses a horizontal sand-filled safety road to stop the car.

The car enters the sand at a speed of 10 m/s and experiences a constant stopping force of 2500 N.

How far does the car travel in the sand before coming to rest?

- A** 2.0 m **B** 4.0 m **C** 20 m **D** 40 m

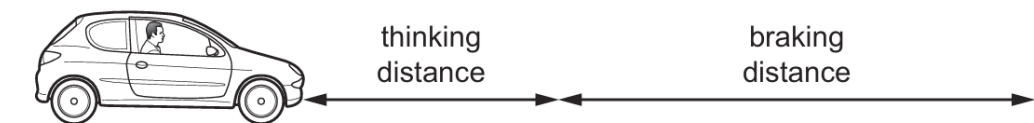
11. O/N/22/11/Q6

What causes the frictional effects experienced by a cyclist to increase?

- A** using a lighter bicycle
B travelling at greater speed
C a smoother tyre surface
D a road surface covered in oil

12. O/N/22/11/Q7

The diagram shows a car at the moment the driver sees a hazard ahead.



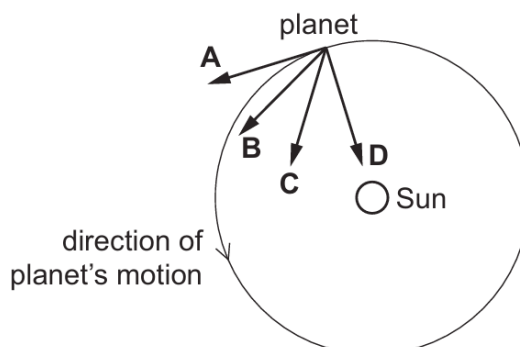
Which expression is equal to the stopping distance?

- A** braking distance + thinking distance
B braking distance – thinking distance
C braking distance $\times$ thinking distance
D braking distance $\div$ thinking distance

13. O/N/22/11/Q8

A planet orbits the Sun. The planet moves in a circular path at constant speed, but is accelerating because it is changing direction.

In which direction does the resultant force act to cause this acceleration?



14. O/N/22/12/Q6

A block of wood is placed on a table. External forces act on the block but the block remains stationary.

A student suggests three conditions for the block to remain stationary.

condition P The resultant force on the block **must** equal zero.

condition Q The resultant moment on the block **must** equal zero.

condition R The external forces **must** act through the centre of mass of the block.

Which conditions are necessary for the block to remain stationary?

A P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

15. O/N/22/12/Q8

A car travels on a horizontal road around a bend at constant speed.

What is the direction of the resultant force on the car?

A backwards

B forwards

C towards the centre of the bend

D towards the road

16. O/N/22/12/Q7

A car travels at 60km/h on a straight road. The road is dry. The driver applies the brakes suddenly. The table shows the thinking distance, the braking distance and the stopping distance.

thinking distance /m	braking distance /m	stopping distance /m
27	22	49

The same car is driven by the same driver at 60km/h on the same road when it is wet. The driver applies the brakes suddenly again.

What is the effect of the wet road on the thinking distance, the braking distance and the stopping distance?

	thinking distance	braking distance	stopping distance
A	no change	increases	increases
B	no change	no change	no change
C	increases	increases	increases
D	increases	no change	increases

17. M/J/22/11/Q2

A driver drives a car at uniform velocity along a road.

The driver sees a hazard and applies the brakes.

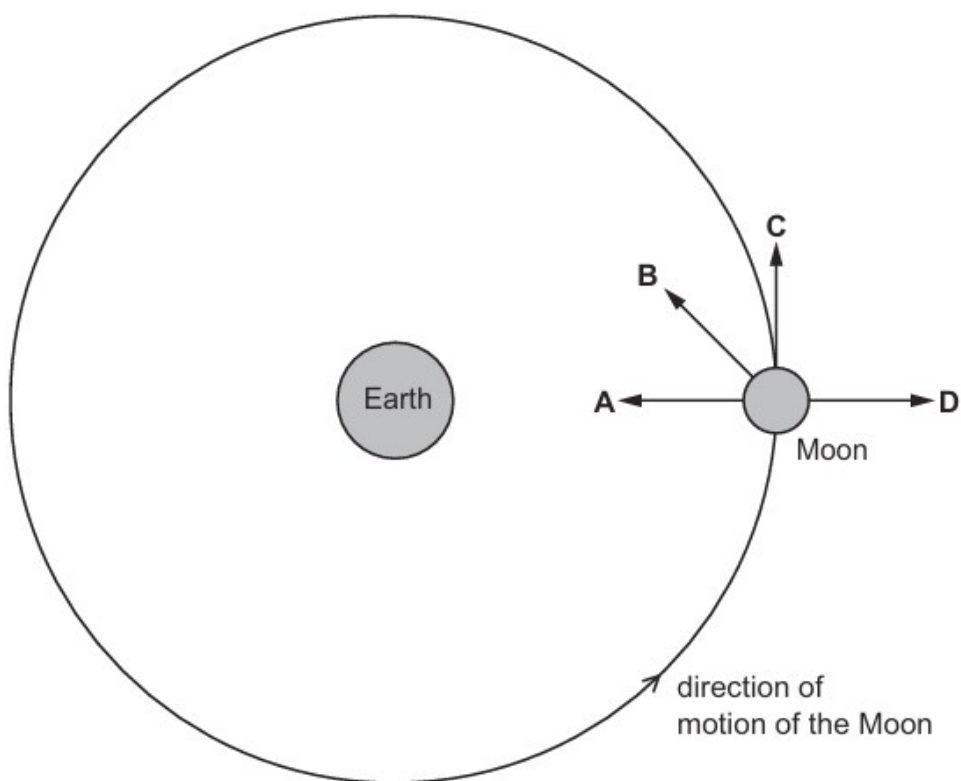
What is the car's motion as it moves through the thinking distance and as it moves through the braking distance?

	motion through the thinking distance	motion through the braking distance
A	deceleration	deceleration
B	deceleration	uniform velocity
C	uniform velocity	deceleration
D	uniform velocity	uniform velocity

18. M/J/22/11/Q3

The diagram represents the Moon in its orbit around the Earth.

Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?



19. M/J/22/12/Q1

A force of 3.0 N and a force of 4.0 N act on an object.

What is the maximum possible resultant of these two forces?

- A** 1.0 N **B** 5.0 N **C** 7.0 N **D** 12 N

20. M/J/22/12/Q4

A 60 kg passenger enters a stationary lift. The gravitational field strength g is 10 N/kg.

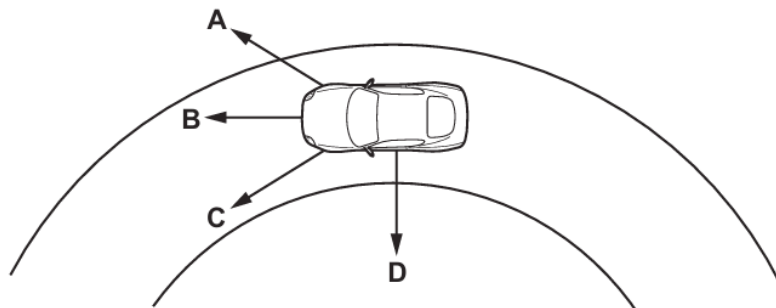
How much force does the floor of the lift exert on the passenger when the lift accelerates upwards at 2.0 m/s^2 ?

- A** 120 N **B** 480 N **C** 600 N **D** 720 N

21. O/N/21/11/Q4

The diagram shows a car going around a circular track at constant speed.

Which arrow shows the direction of the resultant force on the car?



22. O/N/21/11/Q5

An electron has mass.

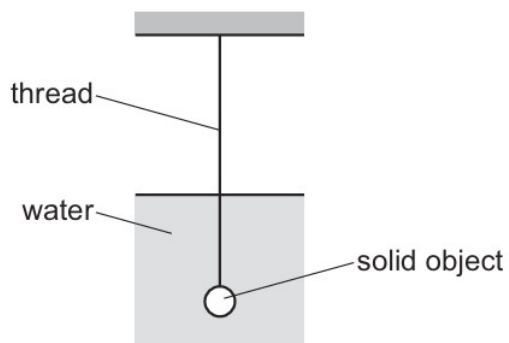
Where does a stationary electron experience a force?

- A** in an electric field only
B in a gravitational field only
C in a gravitational field and an electric field only
D in a gravitational, electric and magnetic field

23. O/N/21/12/Q4

A solid object, immersed in water, hangs from an elastic thread. Three forces act on the object: its weight W , the tension in the thread T , and a force F from the water.

The force F acts upwards.



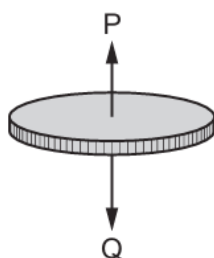
Which equation is correct when the object is stationary?

- A** $F + W = 0$
- B** $F - T = 0$
- C** $F - W - T = 0$
- D** $F - W + T = 0$

24. M/J/21/11/Q4

A coin falls from rest through the air and eventually reaches a constant speed.

There is a resultant force acting on the coin due to the two forces P and Q shown in the diagram.



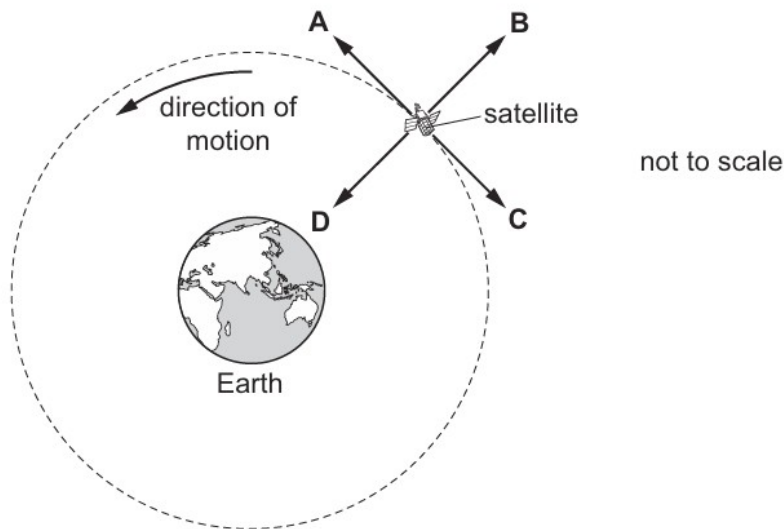
What happens to force P and what happens to the resultant force before the coin reaches constant speed?

	force P	resultant force
A	decreases	increases
B	decreases	decreases
C	increases	decreases
D	increases	increases

25. M/J/21/11/Q5

A satellite is shown moving around the Earth in a circular path at a constant speed.

Which arrow shows the direction of the force on the satellite?



26. M/J/21/12/Q4

A student lists three changes that affect the stopping distance of a car.

- 1 increasing the braking force
- 2 increasing the friction between the tyres and the road
- 3 increasing the speed of the car

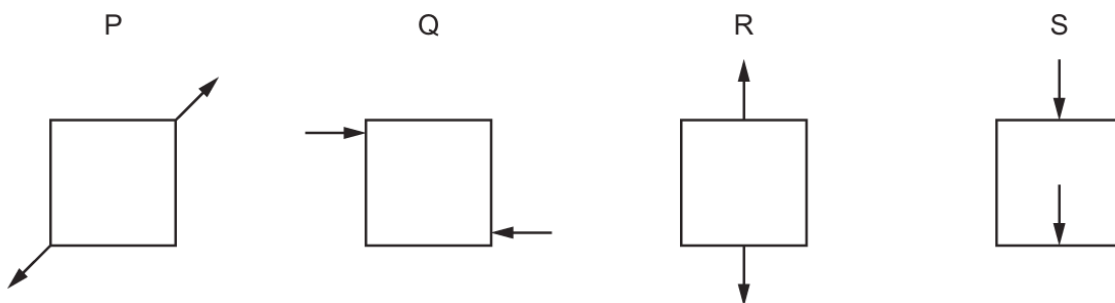
Which change or changes increase the stopping distance?

- A** 1 and 2 **B** 1 and 3 **C** 1 only **D** 3 only

27. M/J/21/12/Q3

The diagrams show the forces acting on four identical solid blocks.

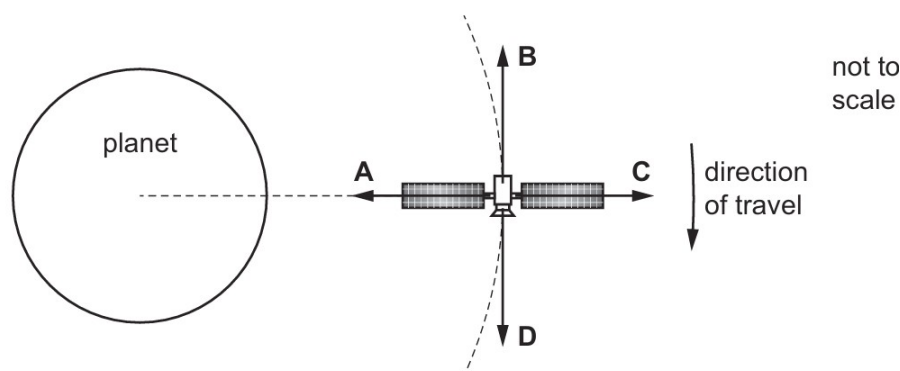
Each arrow represents a force of 20 N.



28. M/J/21/12/Q5

The diagram shows a satellite orbiting a planet at a steady speed.

In which direction does the resultant force act on the satellite?

**29. O/N/20/11/Q3**

A car of mass 800 kg has a forward acceleration of 2.5 m/s^2 . A frictional force of 1200 N opposes the motion of the car.

What is the driving force due to the engine of the car?

- A** 800 N **B** 1200 N **C** 2000 N **D** 3200 N

30. O/N/20/11/Q7

An electrically charged plastic ball is at rest.

Which types of field are caused by the ball?

- A** electric, gravitational and magnetic
B electric and gravitational only
C electric and magnetic only
D gravitational and magnetic only

31. O/N/20/11/Q4

A parachutist falling at a steady speed opens her parachute.

Which row is correct for the direction of the resultant force and for the direction of the acceleration of the parachutist just after her parachute opens?

	resultant force direction	acceleration direction
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

32. O/N/20/11/Q11

Which list contains only quantities that can be changed by a force?

- A mass, shape, velocity
- B mass, shape, volume
- C mass, velocity, volume
- D shape, velocity, volume

33. O/N/20/12/Q4

A car is travelling forwards at high speed. The brakes are applied and the car skids along the road surface.

In which direction and where does the friction force act?

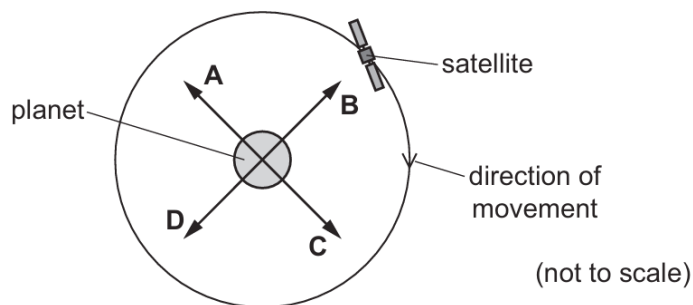
- A backwards on the tyres
- B forwards on the tyres
- C upwards on the tyres
- D downwards on the road

34. O/N/20/12/Q6

A satellite orbits a planet in a circular path as shown. It has constant speed.

There is a force on the planet due to the satellite.

In which direction is the force on the planet?



35. M/J/20/11/Q4

A car of weight 11 000 N moves with constant velocity along a horizontal road. A driving force of 5000 N acts on the car.

What is the force opposing the motion of the car?

- | | | | |
|----------|----------|------------|------------|
| A 5000 N | B 6000 N | C 11 000 N | D 16 000 N |
|----------|----------|------------|------------|

36. M/J/20/11/Q5

A man with an open parachute falls to Earth at constant speed. The following forces act:

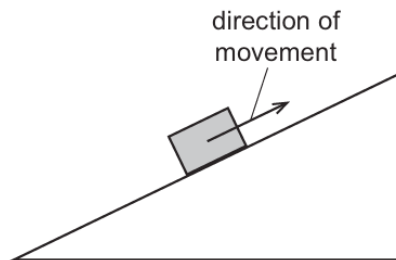
- P the upward force of the parachute on the man
- Q the upward force of the man on the Earth
- R the downward force of the Earth on the parachute
- S the downward force of the man on the parachute

Which two forces are a Newton's third law pair?

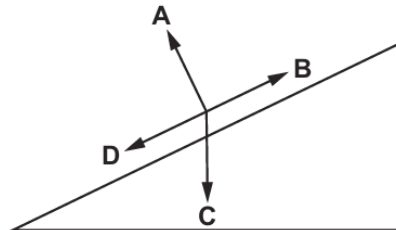
- A** P and Q **B** P and R **C** P and S **D** Q and R

37. M/J/20/11/Q6

A box is pulled up a rough slope, as shown.



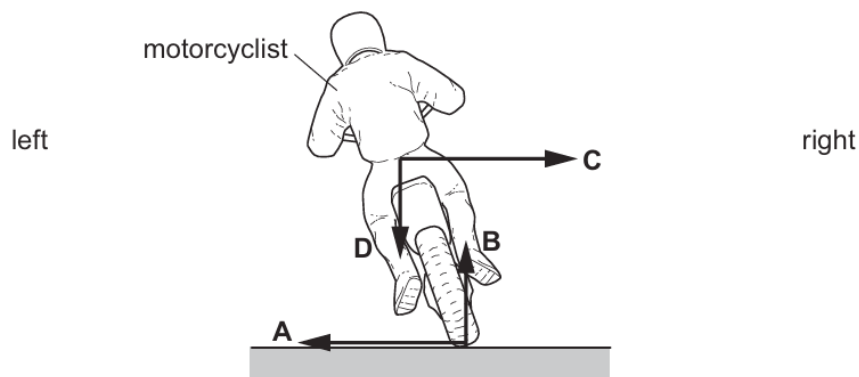
In which direction does friction act on the box?



38. M/J/20/11/Q7

The diagram shows a motorcyclist leaning over in order to turn the corner to the left.

Which force causes him to turn?



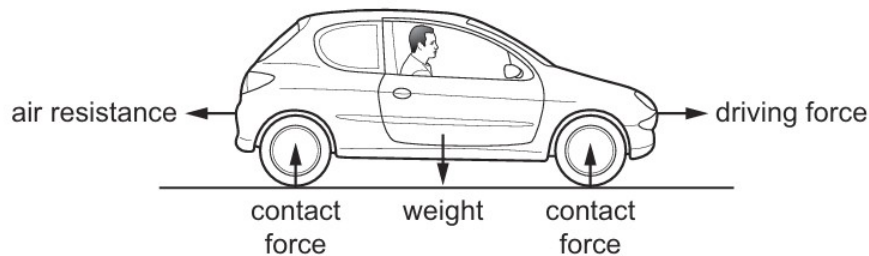
39. M/J/20/11/Q8

Which type of force causes the Earth to orbit the Sun?

- A elastic
- B electrostatic
- C gravitational
- D magnetic

40. O/N/19/11/Q6

A car is accelerating along a straight, horizontal road. The diagram shows forces acting on the car.



Which forces are balanced?

- A contact forces and air resistance
- B contact forces and weight
- C driving force and air resistance
- D driving force and weight

41. O/N/19/11/Q7

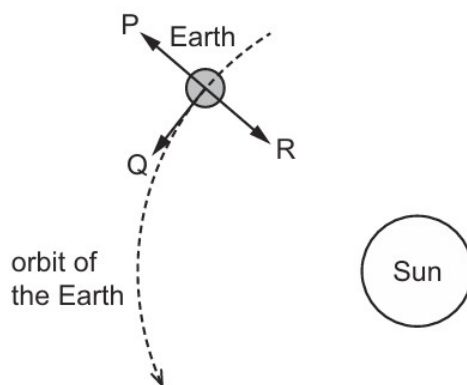
A single force is applied to a body.

What **cannot** happen?

- A a change in the direction of the moving body
- B a change in the mass of the body initially at rest
- C a change in the position of the body initially at rest
- D a change in the speed of the moving body

42. O/N/19/11/Q8

The Earth travels at constant speed in a circular orbit around the Sun.



Which arrows show the direction of the acceleration of the Earth and the direction of the velocity of the Earth?

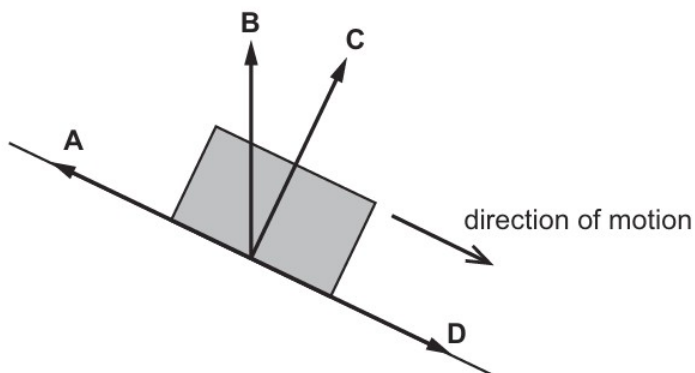
	direction of acceleration	direction of velocity
A	P	Q
B	Q	P
C	Q	R
D	R	Q

43. O/N/19/11/Q9

A body slides down a frictionless slope, as shown.

As the body presses on the surface, the surface pushes back on the body.

In which direction does the surface push back on the body?



44. M/J/19/11/Q5

Four of the gravitational forces that act between bodies in the Solar System are listed below.

- P the force on the Moon due to the Earth
- Q the force on the Earth due to the Sun
- R the force on the Earth due to the Moon
- S the force on the Moon due to the Sun

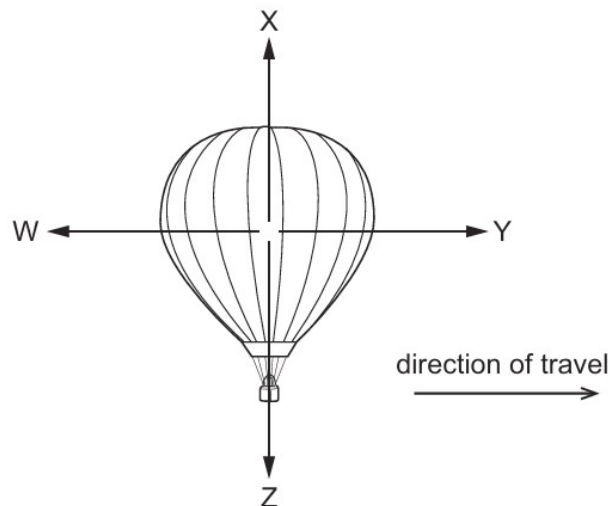
Which two forces are a Newton's third law pair (action and reaction)?

- A** P and Q **B** P and R **C** Q and S **D** R and S

45. M/J/19/11/Q6

A hot-air balloon is travelling at constant velocity and is at a constant height above the ground.

The diagram shows the only four forces acting on the balloon.



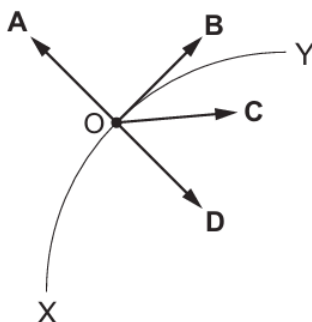
Which statement is correct?

- A** W and Y are equal, X and Z are equal.
- B** W and Y are equal, X is greater than Z.
- C** X and Z are equal, Y is greater than W.
- D** Y is greater than W. X is greater than Z.

46. M/J/19/11/Q7

The diagram represents an object O moving from X to Y along a circular path at constant speed.

What is the direction of the resultant force on O in the position shown?



47. M/J/19/12/Q6

A parachutist of mass 60 kg falls at a steady speed of 10 m/s. The gravitational field strength g is 10 N/kg.

What is the upward force acting on the parachutist?

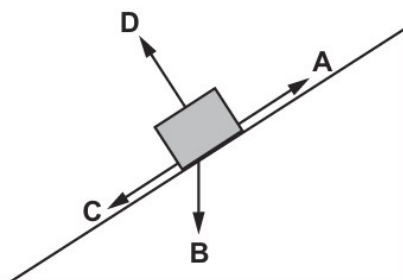
- A** 0 **B** 6.0 N **C** 60 N **D** 600 N

48. M/J/19/12/Q7

A force moves a box up a rough slope.

The diagram shows four forces acting on the box.

Which is the force due to friction on the box?



49. O/N/18/11/Q3

A skydiver is falling at terminal velocity.



Which row describes the acceleration of the skydiver and the velocity of the skydiver?

	acceleration of the skydiver	velocity of the skydiver
A	downwards	constant
B	downwards	zero
C	zero	constant
D	zero	zero

50. O/N/18/12/Q1

An apple falls from a tree.

Which row describes the forces acting on the apple and the motion of the apple as it falls?

	forces acting on the apple	motion of the apple
A	balanced	acceleration
B	balanced	constant speed
C	unbalanced	acceleration
D	unbalanced	constant speed

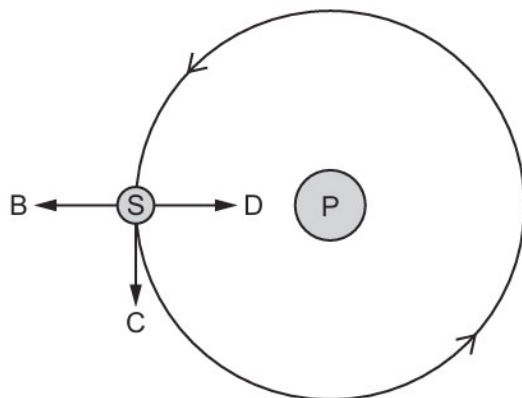
51. O/N/18/12/Q4

On which car is there a resultant force?

- A** a car moving along a straight horizontal road at constant speed
- B** a car moving around a bend at constant speed
- C** a car moving uphill at constant velocity
- D** a car that is stationary

52. M/J/18/11/Q3

The diagram shows a satellite S travelling at a constant speed in a circular orbit around a planet P.



Which statement is correct?

- A** The resultant force on the satellite is zero.
- B** The resultant force on the satellite is in direction B.
- C** The resultant force on the satellite is in direction C.
- D** The resultant force on the satellite is in direction D.

53. M/J/18/11/Q5

The minimum braking distance for a car is tested on a dry road.

The test is then repeated on a wet road.

What happens to the braking distance and to the frictional force between the tyres and the road?

	braking distance	frictional force
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

54. M/J/18/12/Q5

The planets in the Solar System orbit the Sun.

Which statement is correct?

- A** There is a force on each planet away from the Sun.
- B** There is a force on each planet in the direction in which it travels.
- C** There is a force on each planet opposite to the direction in which it travels.
- D** There is a force on each planet towards the Sun.

55. M/J/18/12/Q6

Which forces act on a skydiver who is falling at terminal velocity?

- A** air resistance and weight
- B** air resistance only
- C** weight only
- D** no forces act

56. M/J/18/12/Q11

A force of 1600 N accelerates a car, of mass 800 kg, from rest.

What is the car's acceleration and its velocity after 4.0 s?

	<u>acceleration</u> m/s ²	<u>velocity</u> m/s
A	0.50	0.13
B	0.50	0.50
C	2.0	2.0
D	2.0	8.0

57. O/N/17/11/Q5

A skydiver falls at terminal velocity. He then opens his parachute.

Which row gives the direction of the resultant force on the skydiver and the direction of the acceleration of the skydiver immediately after the parachute opens?

	resultant force	acceleration
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

58. O/N/17/12/Q4

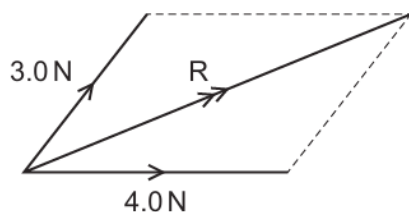
A free-fall skydiver jumps from a plane. As he falls there is a force acting upwards and a force acting downwards on his body. These produce a resultant force.

Before he reaches terminal velocity, how do the sizes of the forces change?

	downward force	upward force	resultant force
A	decreases	decreases	stays the same
B	increases	stays the same	decreases
C	stays the same	increases	decreases
D	stays the same	increases	increases

59. M/J/17/11/Q3

The diagram shows the resultant R of a 3.0 N force and a 4.0 N force acting at a point.



The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90° .

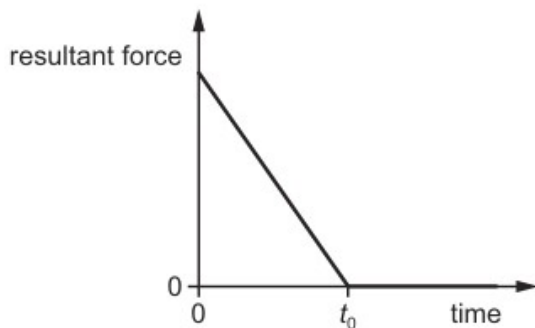
Which value of R is **not** possible?

- A** 4.0 N **B** 5.0 N **C** 6.0 N **D** 7.0 N

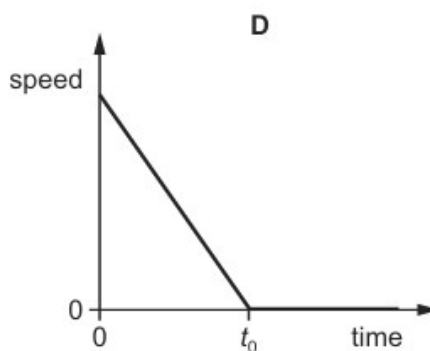
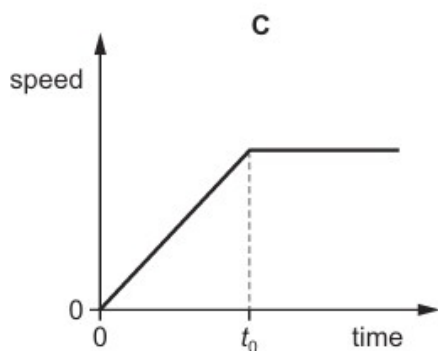
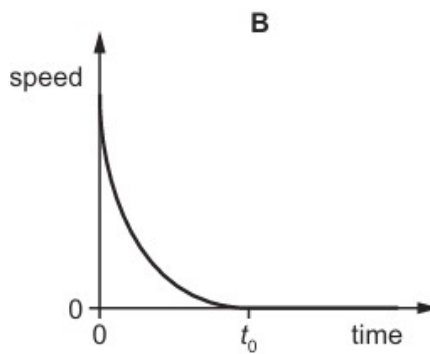
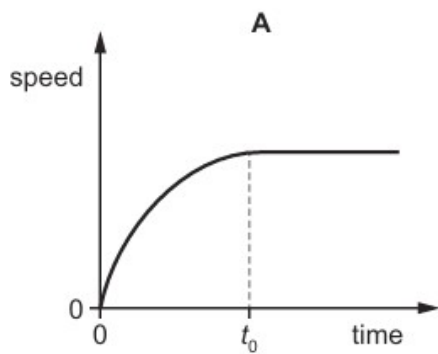
60. M/J/17/11/Q5

A resultant force acts on an object and causes it to move in a straight line.

The graph shows how the resultant force varies with time.



Which graph is the speed-time graph for the object?



61. M/J/17/11/Q6

A skydiver jumps from an aeroplane. After a few seconds, he reaches a terminal velocity without opening his parachute.

Why does he reach terminal velocity?

- A Air resistance becomes greater than his weight and slows him down.
- B Air resistance decreases and he speeds up.
- C Air resistance increases and balances his weight so that his acceleration is zero.
- D His weight decreases and balances the air resistance.

62. M/J/17/11/Q7

A satellite is orbiting the Earth in a circular orbit.

Which two quantities are always in the same direction as each other?

- A the acceleration of the satellite and the displacement of the satellite
- B the acceleration of the satellite and the velocity of the satellite
- C the resultant force on the satellite and the acceleration of the satellite
- D the resultant force on the satellite and the velocity of the satellite

63. M/J/17/11/Q8

A satellite is in a circular orbit around a planet.

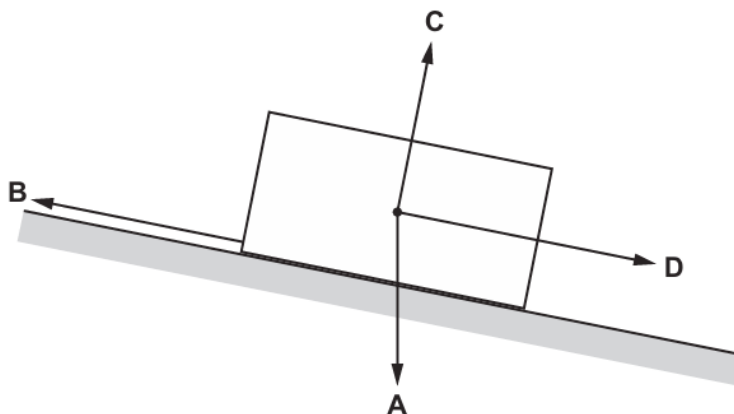
Which statement is correct?

- A Its acceleration is constant in direction but not in size.
- B Its acceleration is constant in size but not in direction.
- C Its gravitational potential energy varies.
- D Its velocity is constant.

64. M/J/17/11/Q9

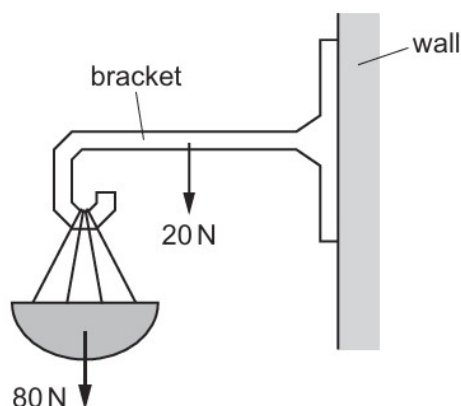
The diagram shows a block of wood resting on a sloping board.

Which arrow shows the direction of the gravitational force acting on the block?



65. M/J/17/12/Q6

A hanging basket is fixed to a wall by a bracket.



The weight of the basket is 80 N. The weight of the bracket is 20 N.

What is the size of the upwards force exerted on the bracket by the wall?

- A** 60 N **B** 70 N **C** 90 N **D** 100 N

66. M/J/17/12/Q7

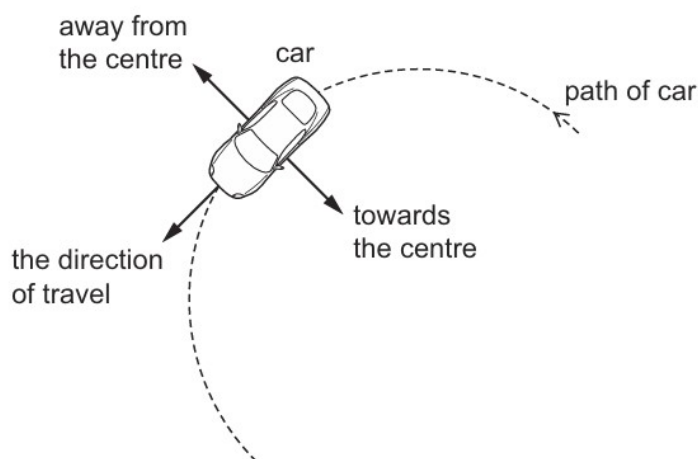
A train of mass 240 000 kg is travelling at a speed of 60 m/s. The brakes are applied and it decelerates for 10 minutes until it comes to rest.

What is the average resultant force?

- A** 24 000 N **B** 40 000 N **C** 480 000 N **D** 720 000 N

67. M/J/17/12/Q8

A car travels at a constant speed along a circular, horizontal path.



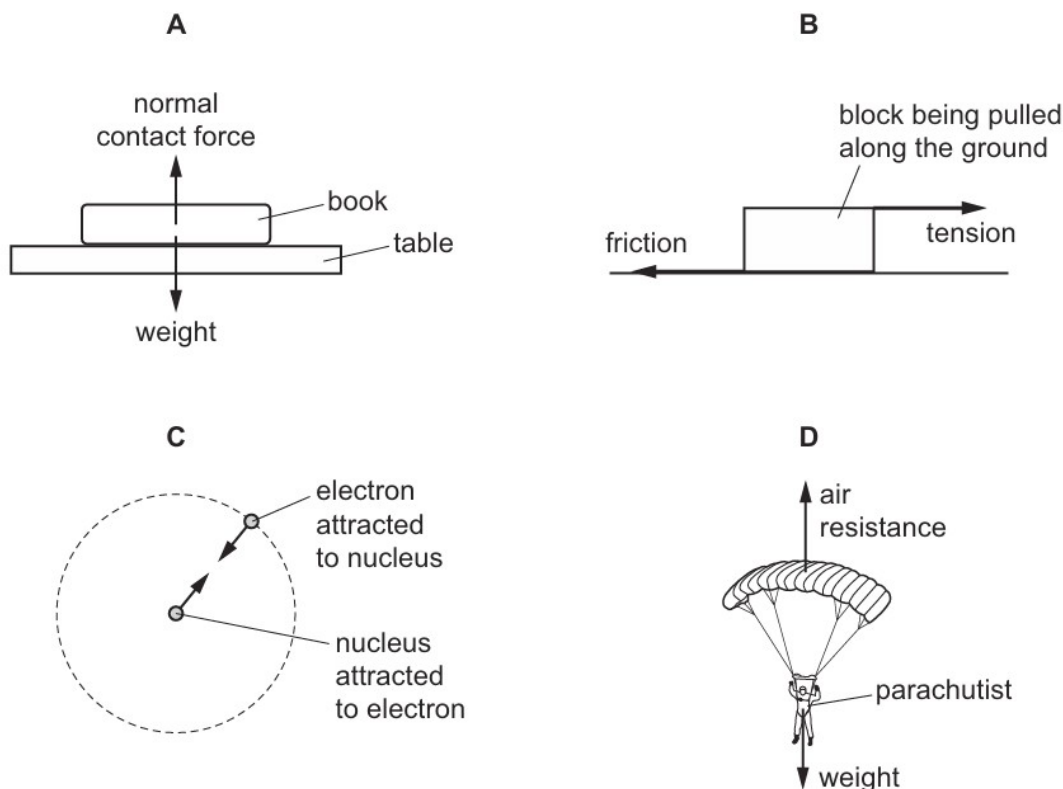
Which statement describes the forces acting on the car?

- A** They are balanced as the car is moving at constant speed.
B They are unbalanced with a resultant in the direction away from the centre.
C They are unbalanced with a resultant in the direction of travel of the car.
D They are unbalanced with a resultant in the direction towards the centre.

68. O/N/16/11/Q6

Each diagram shows two forces that are equal in magnitude.

Which diagram shows two forces that are a Newton's third law pair of forces?



69. O/N/16/12/Q3

Two aluminium spheres P and Q, of masses 0.50 kg and 1.0 kg respectively, are released at the same time from the same height.

Why do they have the same acceleration?

- A** The density of P is equal to the density of Q.
- B** The force on P is equal to the force on Q.
- C** The force on Q is double the force on P.
- D** The gravitational field strength for Q is double that for P.

70. O/N/16/12/Q5

A force of 20 N acts on an object of mass 5.0 kg in the forward direction. A second force also acts on the object. The forward acceleration of the object is 3.0 m/s^2 .

What is the size and direction of the second force?

	size of force / N	direction of force
A	5.0	backwards
B	15	backwards
C	15	forwards
D	35	forwards

71. M/J/16/11/Q5

A man pulls a sledge of mass 25 kg across level ground with a horizontal force of 60 N. A constant force of friction of 20 N acts on the sledge.

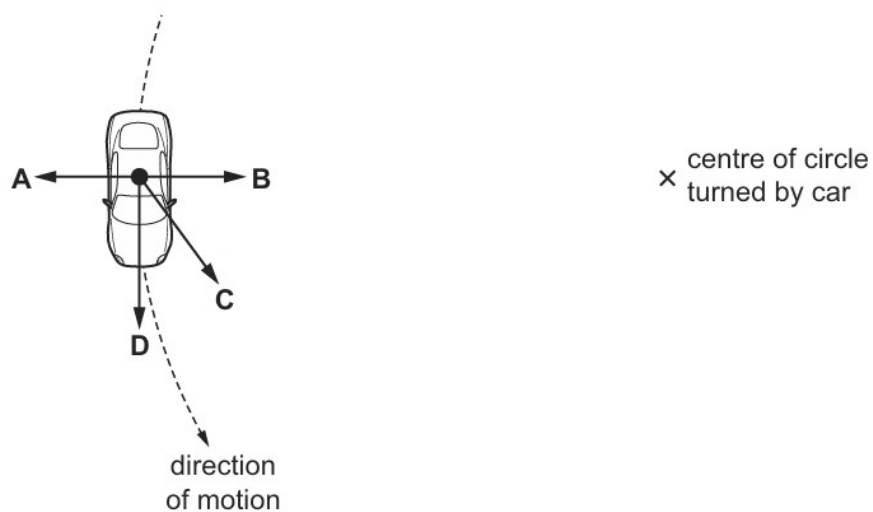
What is the acceleration of the sledge?

- A** 0.63 m/s^2 **B** 1.6 m/s^2 **C** 2.4 m/s^2 **D** 3.2 m/s^2

72. M/J/16/11//Q6

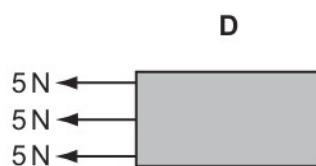
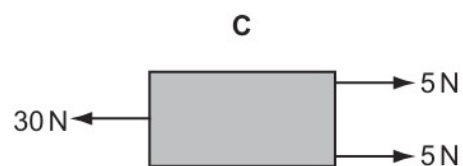
A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?



73. O/N/15/11/Q4

Which object has the largest resultant force?



74. O/N/15/11/Q5

In which example does friction act in the direction of forward motion of the object on which it acts?

- A** on a suitcase accelerating on a conveyor belt
- B** on sandpaper rubbing a wooden block
- C** on the skis of a skier accelerating downhill on smooth snow
- D** on the tyres of a braking car

75. O/N/15/12/Q5

An apple of mass 0.15 kg and weight 1.5 N falls from a tree. At one point during its fall, the air resistance on the apple is 0.60 N upwards.

What is the acceleration of the apple at this point?

- A** 4.0 m/s²
- B** 6.0 m/s²
- C** 10 m/s²
- D** 14 m/s²

76. O/N/15/12/Q6

At time t_1 , a stone is dropped from a stationary balloon. The stone reaches terminal velocity at time t_2 .

Which row gives the acceleration of the stone at time t_1 and at time t_2 ?

	<u>acceleration at t_1</u> m/s ²	<u>acceleration at t_2</u> m/s ²
A	0	0
B	10	0
C	0	10
D	10	10

77. O/N/15/12/Q7

Four of the gravitational forces that act between bodies in the Solar System are described below.

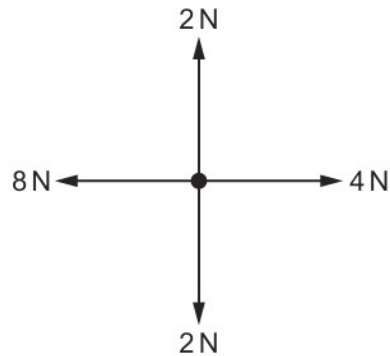
- P** the force on the Moon due to the Earth
- Q** the force on the Earth due to the Sun
- R** the force on the Earth due to the Moon
- S** the force on the Moon due to the Sun

Which two forces are a Newton's third law pair (action and reaction)?

- A** P and Q
- B** P and R
- C** Q and S
- D** R and S

78. M/J/15/11/Q6

Four forces act at a point as shown.



What is the size of the resultant force?

- A** 0 N **B** 4 N **C** 6 N **D** 8 N

79. M/J/15/11/Q9

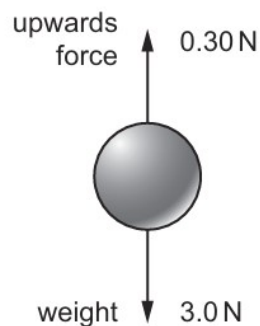
Newton's third law involves two quantities which are equal in size and opposite in direction.

What is the unit for these two quantities?

- A** J
B m/s^2
C N
D W

80. M/J/15/12/Q5

A metal ball of mass 0.30 kg and weight 3.0 N is held so that it is below the surface of oil. It experiences an upwards force of 0.30 N.



When the ball is released, what is its initial acceleration?

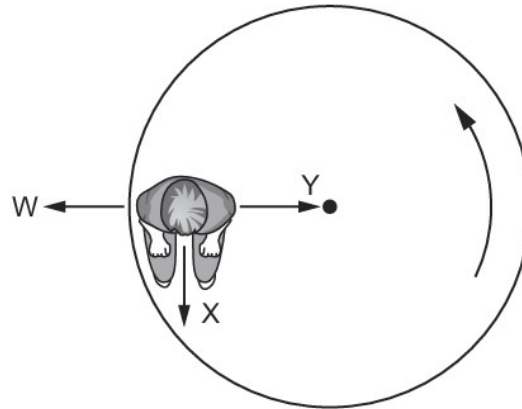
- A** 1.0 m/s^2 **B** 9.0 m/s^2 **C** 10 m/s^2 **D** 11 m/s^2

81. O/N/14/11/Q5

A boy sits on a playground roundabout (carousel).

The roundabout carries the boy in a horizontal, anticlockwise circle at a constant speed.

The diagram shows the view of the roundabout from above.



What describes the resultant force on the boy?

- A The resultant force is zero.
- B The resultant force is in direction W.
- C The resultant force is in direction X.
- D The resultant force is in direction Y.

82. O/N/14/12/Q1

When a heavy coin falls a short distance towards the ground it does **not** reach terminal velocity.

Why is this?

- A The coin has not hit the ground.
- B The weight of the coin equals the air resistance.
- C The weight of the coin increases as air resistance increases.
- D The weight of the coin is always more than air resistance.

83. O/N/14/12/Q2

A car travels along a road at 50 km/h.

The driver applies the same braking force at the same place on a day when the surface is dry and then on a day when the road is wet.

On the **wet** surface, how many of these distances are greater than on the dry surface?

braking distance

stopping distance

thinking distance

- A 0 B 1 C 2 D 3

84. M/J/14/11/Q4

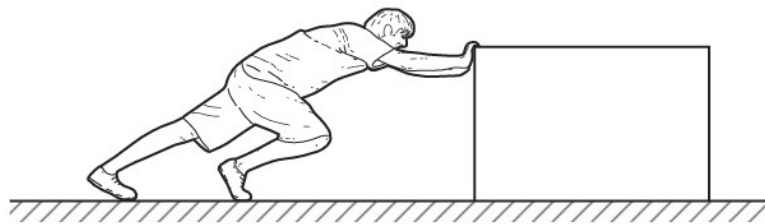
An object falls from rest through the air and the air resistance on it increases. The object reaches terminal velocity.

Which quantity **decreases** until its terminal velocity is reached?

- A acceleration
- B kinetic energy
- C speed
- D weight

85. M/J/14/11/Q5

A man pushes a heavy box along the ground.



A force acts between the man's hands and the box.

Another force acts between the man's feet and the floor.

In which direction do these forces act on the man?

	force on man's hands	force on man's feet
A	towards the left	towards the left
B	towards the left	towards the right
C	towards the right	towards the left
D	towards the right	towards the right

86. M/J/14/11/Q6

In which situation is the resultant force on the body equal to zero?

- A a car turning a corner at constant speed
- B a rock falling freely on the Moon
- C a train going up a straight hill at constant speed in a straight line
- D an aircraft accelerating along a runway in a straight line

87. M/J/14/11/Q7

A satellite is orbiting the Earth.

What is the direction of the force on the satellite causing this circular motion?

- A** away from the centre of the Earth
- B** in the direction of motion of the satellite
- C** in the opposite direction to the motion of the satellite
- D** towards the centre of the Earth

88. M/J/14/11/Q11

A force is applied to a body.

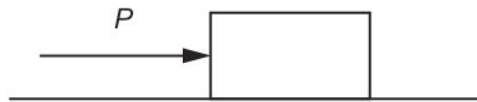
Which property of the body **cannot** be changed by the force?

- A** its mass
- B** its shape
- C** its size
- D** its velocity

89. M/J/14/12/Q5

The diagram shows a block of stone on a rough horizontal surface.

Force P acts on the block as shown.



The block is at rest. A frictional force F acts on the block.

Which row shows the direction and size of F ?

	direction of F	size of F
A	to the left	more than P
B	to the right	more than P
C	to the left	same as P
D	to the right	same as P

90. O/N/13/11/Q4

A student kicks a ball horizontally along the ground. As he does so, his foot applies a force to the ball.

At the same time, the ball applies a force to his foot.

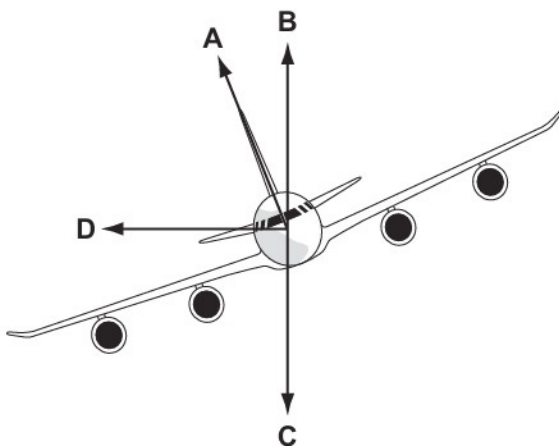
How do these forces compare?

- A** The force on the foot is equal to the force on the ball and in the opposite direction.
- B** The force on the foot is equal to the force on the ball and in the same direction.
- C** The force on the foot is smaller than the force on the ball and in the opposite direction.
- D** The force on the foot is smaller than the force on the ball and in the same direction.

91. O/N/13/11/Q5

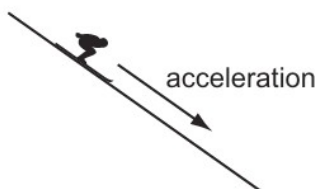
The diagram shows an aeroplane turning in a horizontal circle at constant speed.

In which direction is there a resultant force?



92. O/N/13/12/Q4

A skier of mass 80 kg accelerates down a slope at 2 m/s^2 .



The gravitational field strength g is 10 N/kg .

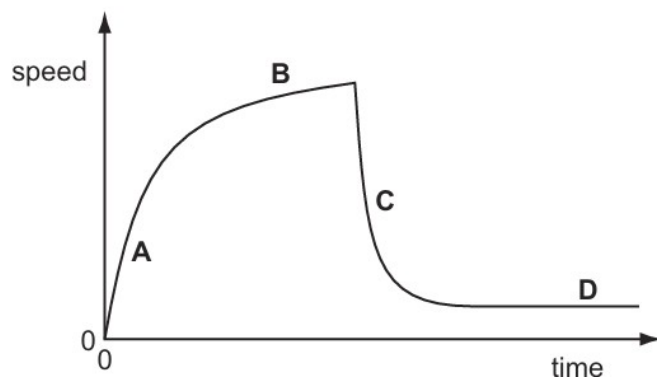
What is the weight of the skier and the resultant force on the skier?

	weight / N	resultant force / N
A	8	168
B	80	240
C	800	160
D	800	640

93. M/J/13/11/Q3

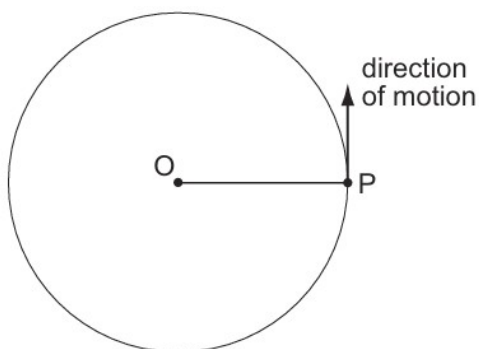
The speed-time graph for a falling skydiver is shown below. As he falls, the skydiver spreads out his arms and legs and then opens his parachute.

Which part of the graph shows the skydiver falling with terminal velocity?



94. M/J/13/11/Q4

A particle P is moving in a horizontal circle about O. P moves at constant speed.



Which statement is true?

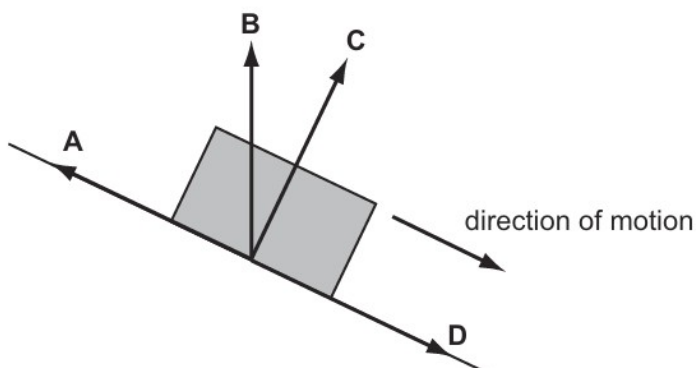
- A** A force of constant size acts on P in the direction of motion.
- B** A force of constant size acts on P towards O.
- C** The force on P varies in size as it moves around the circle.
- D** There is no resultant force acting on P.

95. M/J/13/11/Q5

A body slides down a frictionless slope as shown.

As the body presses on the surface, the surface pushes back on the body. The force of the surface on the body is sometimes called the reaction force.

In which direction does the reaction force act?

**96. M/J/13/12/Q6**

Balanced forces are acting on a moving body.

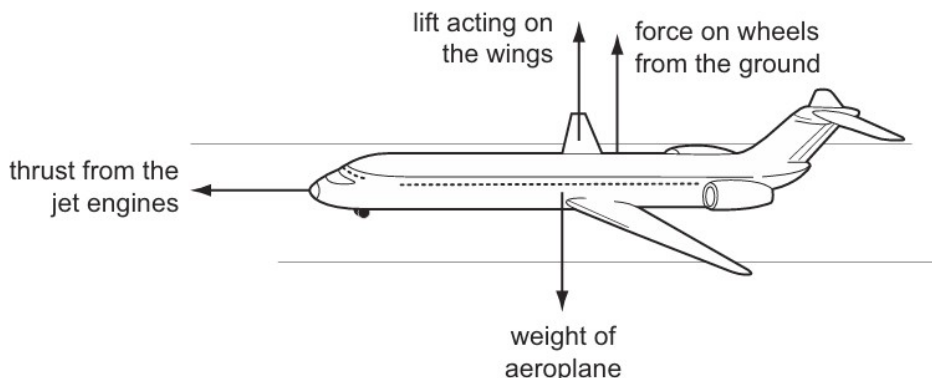
What happens to the direction of movement and to the speed of the body?

	direction of movement	speed
A	changes	changes
B	changes	does not change
C	does not change	changes
D	does not change	does not change

97. O/N/12/11/Q5

An aeroplane is half way along a runway before it takes off.

The arrows show the directions of four forces acting on the aeroplane.



The drag forces on the aeroplane are negligible.

The gravitational field strength g is 10 N/kg .

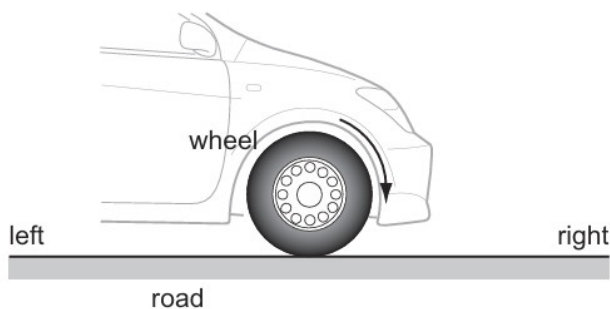
The acceleration of the aeroplane is 2 m/s^2 .

What is the largest force?

- A** the force on the wheels from the ground
- B** the lift acting on the wings
- C** the thrust from the jet engines
- D** the weight of the aeroplane

98. O/N/12/11/Q6

The front wheel of a car is turned in a clockwise direction by the engine as the car accelerates towards the right, as shown in the diagram. There is a force of friction between the wheel and the road.

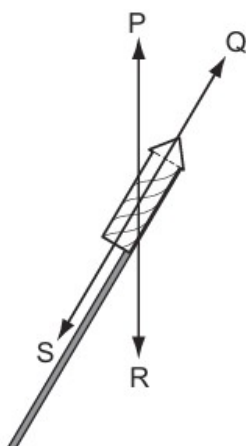


What are the directions of the frictional forces on the wheel of the car and on the road?

	direction of frictional force on wheel	direction of frictional force on road
A	to the left	to the left
B	to the left	to the right
C	to the right	to the left
D	to the right	to the right

99. O/N/12/12/Q5

The diagram shows a firework rocket.



As the rocket flies through the air, three forces act on it. These forces are weight, thrust and air resistance.

What are the three forces?

	thrust	air resistance	weight
A	P	R	S
B	P	S	R
C	Q	R	S
D	Q	S	R

100. O/N/12/12/Q6

An object has a mass of 15 kg. It is pushed horizontally by a force of 40 N. The frictional force is 10 N.

What is the acceleration of the object?

- A** 0.50 m/s^2 **B** 1.5 m/s^2 **C** 2.0 m/s^2 **D** 2.7 m/s^2

101. M/J/12/11/Q5

A hard stone hits the ground and comes to rest almost immediately.

As the stone hits the ground, what is the direction and the size of the force acting on the ground?

- A** downwards and equal to the weight of the stone
- B** downwards and larger than the weight of the stone
- C** upwards and equal to the weight of the stone
- D** upwards and larger than the weight of the stone

102. M/J/12/11/Q6

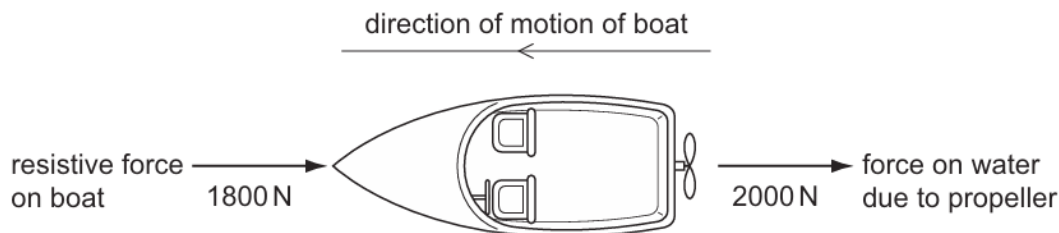
A car of mass 1500 kg is towing a trailer of mass 1100 kg along a level road. The acceleration of the car is 1.30 m/s^2 .

Ignoring friction and air resistance, what is the driving force on the car?

- A** 1430 N **B** 1950 N **C** 2000 N **D** 3380 N

103. O/N/11/11/Q4

The propeller on a boat pushes water backwards with a force of 2000 N. The boat moves through the water against a total resistive force of 1800 N.



According to Newton's third law, what is the forward force on the propeller due to the water?

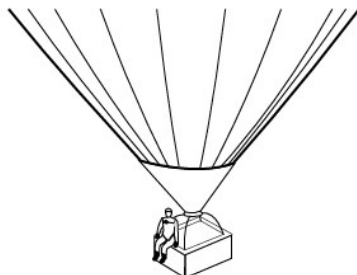
- A** 200 N **B** 1800 N **C** 2000 N **D** 3800 N

104. O/N/11/11/Q5

The diagrams show a parachutist in four positions after she jumps from a high balloon.

At which position does she have terminal velocity?

- A** not yet moving



- B** accelerating



- C** not accelerating



- D** not moving

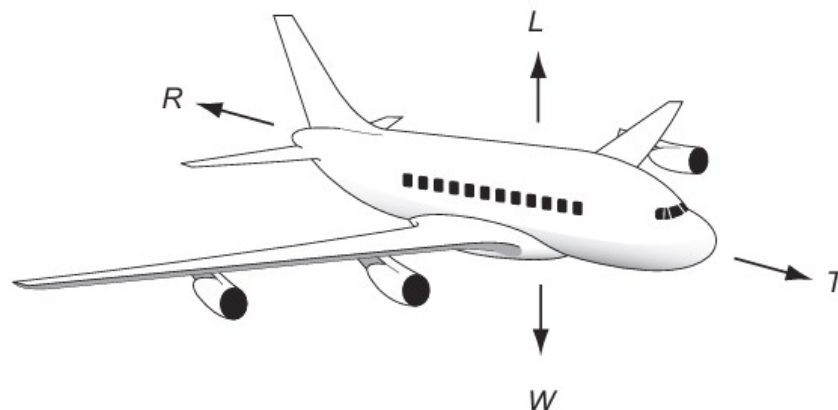


105. O/N/10/11/Q5

An aircraft, flying at a constant height, is gaining speed.

The four forces acting are

- L lift due to the wings
- R air resistance
- T the thrust due to the engines
- W the weight



Which row is correct?

	vertical forces	horizontal forces
A	$L = W$	$T = R$
B	$L > W$	$T > R$
C	$L = W$	$T > R$
D	$L > W$	$T = R$

106. O/N/10/11/Q6

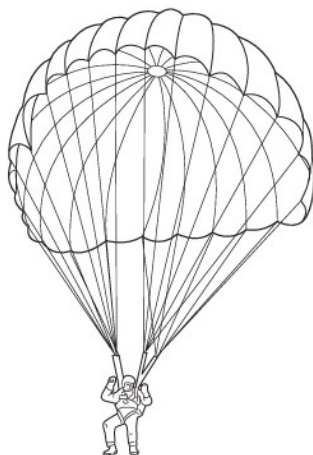
A car travels along a road. The driver stops the car by pushing his foot down on the brake pedal.

What does **not** change if he pushes harder on the brake pedal?

- A** the braking distance
- B** the braking force
- C** the stopping distance
- D** the thinking distance

107. M/J/10/11/Q5

Two men jump out of an aeroplane at the same time. One of the men opens his parachute and the other man remains in free-fall.



Why is the man in free-fall moving faster than the parachutist?

- A** The man in free-fall experiences greater air resistance.
- B** The man in free-fall has a greater mass.
- C** The parachutist experiences greater air resistance.
- D** The parachutist has not reached terminal velocity.

108. O/N/09/Q5

A tractor pulls a trailer at a constant speed.

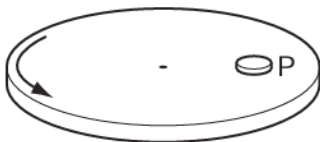
The tractor exerts a forward force of 1600 N on the trailer.

What is the force exerted by the trailer on the tractor?

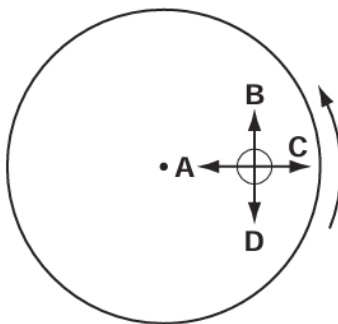
- A** 0 N
- B** 1600 N backwards
- C** 1600 N forwards
- D** 3200 N forwards

109. O/N/09/Q6

A turntable rotates at constant speed. A coin is placed on the turntable at P. The friction force between the coin and the turntable keeps the coin in the same position on the turntable.

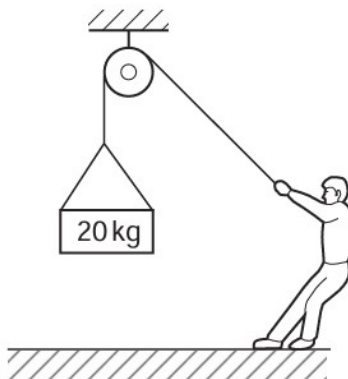


In which direction does the friction force act?



110. M/J/09/Q5

A person just supports a mass of 20 kg suspended from a rope.



What is the resultant force acting on the mass?

- A** 0 N **B** 10 N **C** 20 N **D** 200 N

111. M/J/08/Q5

What keeps an electron moving in a circle around the nucleus of an atom?

- A** a gravitational force away from the nucleus
- B** a gravitational force towards the nucleus
- C** an electrostatic force away from the nucleus
- D** an electrostatic force towards the nucleus

112. O/N/07/Q2

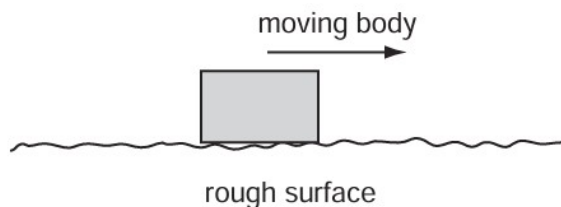
Forces of 30 N and 50 N act on the same body, but in different directions.

Which value could **not** be the resultant force on the body?

- A** 10 N **B** 30 N **C** 50 N **D** 70 N

113. O/N/07/Q4

When a body moves across a rough surface, a frictional force is produced.



Which statement about this force is **always** true?

- A** It acts in the direction of the motion.
B It is equal in value to the force producing the motion.
C It makes the body recoil in the opposite direction after stopping it.
D It opposes the motion across the surface.

114. O/N/06/Q4

When a block of wood of mass 2 kg is pushed along the horizontal flat surface of a bench, the friction force is 4 N.

When the block is pushed along the bench with a force of 10 N, it moves with a constant

- A** speed of 3 m/s.
B speed of 5 m/s.
C acceleration of 3 m/s².
D acceleration of 5 m/s².

115. M/J/06/Q2

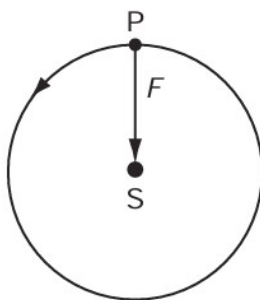
A force of 20 N pushes an object of mass 5.0 kg along a rough horizontal surface where the frictional force is 5.0 N.

What is the acceleration of the object?

- A** 1.0 m/s² **B** 2.0 m/s² **C** 3.0 m/s² **D** 4.0 m/s²

116. O/N/05/Q5

A body P moves in a circle around a point S . A force F keeps it moving in the circle.

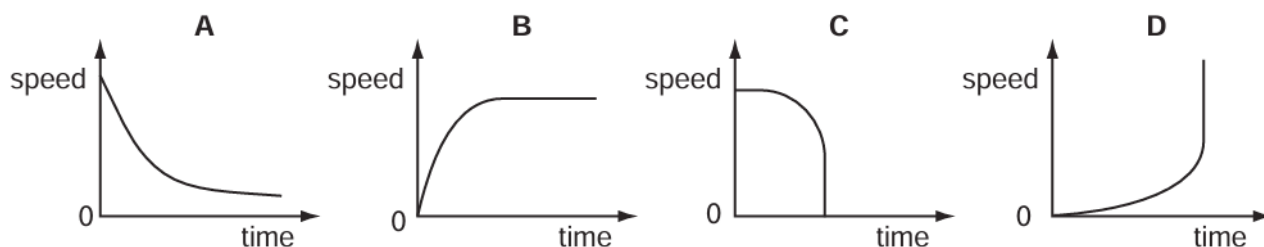


What happens if the force F suddenly disappears?

- A** P moves directly towards S .
- B** P moves in a circle closer to S .
- C** P moves away from S in a curved path.
- D** P goes off in a straight line.

117. M/J/05/Q2

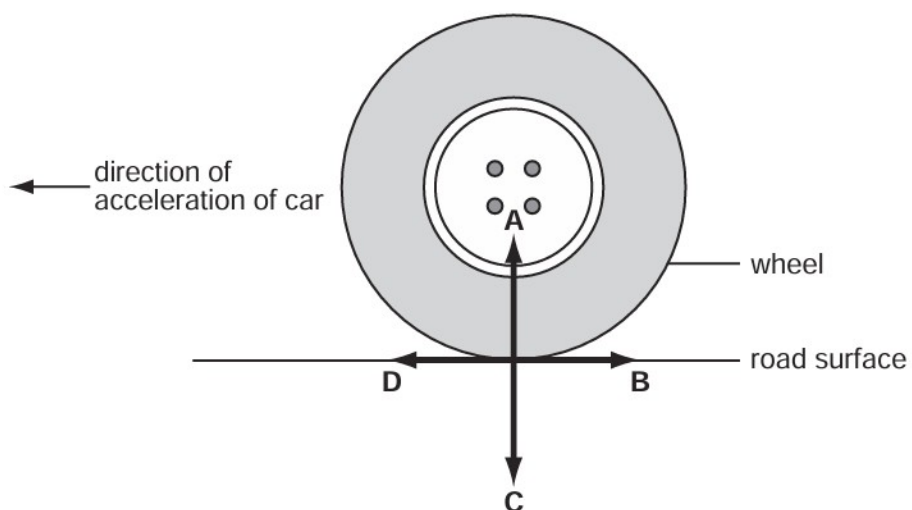
Which graph represents the motion of a body falling vertically that reaches a terminal velocity?



118. O/N/04/Q5

The wheel of a moving car is driven by the engine. The car is accelerating in the direction shown.

In which direction does the frictional force act on the wheel?



119. M/J/04/Q6

Which is a statement of Newton's third law of motion?

- A** Every force causes a reaction.
- B** If there is no resultant force on a body then there is no acceleration.
- C** The forces acting on a body are always equal and opposite.
- D** To every action there is an equal but opposite reaction.